MATHAN PONRAJ

AI Backend Engineer | Python AI Developer

Seeking a Python Developer role in Bengaluru | Immediate Joiner

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PROFESSIONAL SUMMARY

Python AI and backend developer based in Bengaluru with hands-on experience building REST APIs with FastAPI, GenAI and RAG applications with LangChain, and ETL data pipelines. Has taken 3+ projects from raw data to live deployment. Immediate joiner seeking a full-time Python Developer role in Bengaluru.

TECHNICAL SKILLS

AI & LLM: LangChain, RAG, LLM APIs (OpenAI, Gemini), Prompt Engineering, Hugging Face, ChromaDB, AI Agents
Backend: Python, FastAPI, Flask, REST APIs, SQL, PostgreSQL, SQLite, Git, GitHub, Postman
Frontend: HTML, CSS, JavaScript, React.js
Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, ETL, Star Schema, Power BI, Excel
Deployment: Render, Vercel

PROFESSIONAL EXPERIENCE

Freelance Developer & Team Lead **Ongoing (1+ year)**

- Lead a 3-member technical team delivering web applications that combine backend engineering and AI components.
- Built and deployed static and dynamic web pages; completed and hosted 3+ live production applications.
- Handled end-to-end Instagram content creation, including professional video editing and Photoshop design.

Data Science Intern | Rinex.ai **Mar 2025 – May 2025**

PROJECTS

TalentIQ AI | Python, FastAPI, LangChain, RAG, React.js, Vite, Gemini API, Vercel, Render

- Built a GenAI career-prep app that analyses an uploaded resume against a target role to surface skill gaps and suggest resume improvements using NLP.
- Developed the FastAPI backend and a LangChain RAG pipeline with semantic chunking so related skills stay together in embeddings; added mock interviews at multiple difficulty levels.

TrendTube – YouTube Trending Data Pipeline | Python, PostgreSQL, ETL, Pandas, Psycopg, YouTube Data API v3

- Built a Python ETL pipeline that pulls trending videos across 9 regions from the YouTube Data API, cleans and validates them with Pandas, and loads them into PostgreSQL.
- Designed a star schema (dim_video, dim_region, dim_date, fact_trending) and an engagement metric to compare regional trends with plain SQL.

EduHelpDesk – Academic Support Ticketing System | Python, Flask, SQLite, REST APIs, RBAC, Render

- Developed a role-based ticketing platform where students raise tickets, staff resolve them and managers track progress on separate dashboards; deployed on Render.
- Implemented SLA tracking that measures ticket age against deadlines, flags overdue cases and keeps a full history of every status change.

LapPick – Laptop Price & Recommendation | Python, XGBoost, Scikit-learn, Streamlit GitHub | Live Demo

- Trained an XGBoost model on about 1,300 laptop listings to predict price from specifications ($R^2 \approx 0.91$) after outlier cleaning and category encoding in a Scikit-learn pipeline.
- Built a Streamlit app that recommends laptops by budget, RAM, CPU and storage filters.

EDUCATION

M.E. in Computer Science & Engineering | Karpagam Academy of Higher Education, Coimbatore | First Class **2024 – 2026**

B.E. in Computer Science & Engineering | Government College of Engineering, Bodinayakkanur | First Class **2020 – 2024**

CERTIFICATIONS

- Data Science and Machine Learning using Python
- Deloitte Virtual Data Analytics Internship
- Advanced SQL and Excel for Data Analysis
- Python Programming